

## Uni-Fi: a short guide to the radio mesh

**Uni-Fi** is MeshUSF's emergency-first messaging prototype for Heltec V3 LoRa radios. Its purpose is to let nearby radios exchange short messages directly, including a distress signal and a human acknowledgement, without depending on a phone, Wi-Fi, cellular service, or a central server.

### The idea in one minute

Each Heltec is a small independent radio computer. It listens for nearby LoRa packets, checks messages it can decrypt, and may relay eligible packets for other nodes. A phone can connect over Bluetooth or USB to make configuration and reading messages easier. The phone does not carry the mesh traffic.

Person presses button



Heltec builds a short Uni-Fi event



MeshCore puts it in a radio packet and encrypts the group message



LoRa carries it directly or through other Heltec relays



Another Heltec decrypts it, updates its local state, and alerts its user



That user may press the button to send a human acknowledgement back

The network has no automatic guarantee of delivery. The node can report that it queued a message; a matching acknowledgement is evidence another node received the event and someone chose to respond.

### MeshCore and Uni-Fi: who does what?

| MeshCore provides                                   | Uni-Fi adds                                                                    |
|-----------------------------------------------------|--------------------------------------------------------------------------------|
| LoRa packet format and radio scheduling             | Emergency message types such as distress, medical, pickup, and acknowledgement |
| Encrypted group messages and direct messages        | Rules for recognizing Uni-Fi events inside group messages                      |
| Node identities and public signed adverts           | Button and LED behavior for distress and response                              |
| Packet forwarding, duplicate filtering, and routing | A locked product radio profile and a simpler companion interface               |

Uni-Fi is a focused product built on MeshCore; it does not replace MeshCore's radio network. The current firmware keeps MeshCore's established packet and routing machinery while selecting a smaller Heltec V3 build for this project.

### How a message travels

1. A person presses the Heltec's USER button, or a connected app asks the firmware to send an event.

2. Uni-Fi creates text describing the event. It includes a type and event ID so nodes can recognize distress and match a reply to the right event.
3. MeshCore wraps that text in a group-message packet, encrypts it with the shared Uni-Fi channel key, and schedules it for LoRa transmission.
4. Nearby compatible nodes receive the packet. Eligible nodes can relay it; relays suppress duplicates and the Uni-Fi build limits flood forwarding.
5. A recipient with the channel key verifies and decrypts the group message. Its Uni-Fi firmware updates its short-lived local presence/distress state, shows a preview, and signals a newly received distress with two long LED blinks.
6. A person can hold USER to send a new encrypted group message referring to the original event ID. The first node recognizes only a matching reply to its own outstanding event.

The two long blinks mean that a distress was received or that a matching reply arrived. The display provides the context. Three short blinks indicate that the local node accepted a distress for transmission; they do not prove that it reached another radio.

## Does the node work without a phone?

Yes, when it is powered and within radio range of another compatible node, directly or through relays. The firmware runs the receive, decrypt, display, alert, and relay steps on the Heltec itself. It can send the current button distress and a button acknowledgement with no app connected. It also sends a periodic presence message.

The phone is a nearby companion. It can set the saved node name and time, display messages and presence, and offer more sending controls. The current firmware keeps a bounded queue of up to 64 received frames for a companion to retrieve later. That queue and short-lived presence/distress records are held in memory and do not survive a power restart. The screen is useful for alerts and previews, but is not yet a persistent, searchable inbox.

## What does “presence” tell us?

The node periodically broadcasts an encrypted Uni-Fi presence message with its saved name and node identifier. Other nodes can keep a local record and refresh its expiry when they hear another message. This is not a GPS fix.

GPS is disabled in the current prototype, and the public MeshCore advert does

not include GPS coordinates. Therefore the current build cannot locate nodes on a map or broadcast their current positions.

## What is implemented, and what comes later?

| In the current prototype                                                  | Later work                                                              |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Heltec V3 firmware build for the selected 910.525 MHz radio profile       | GPS module, location updates, and a verified map flow                   |
| Encrypted group channel and eligible multi-hop relaying                   | Persistent on-device message history                                    |
| Button distress and button human acknowledgement                          | More physical quick-message choices and usability testing               |
| LED alerts, OLED previews, periodic presence, and temporary local records | Retry policy, stronger sender identity binding, and key lifecycle tools |
| BLE/USB companion and lightweight browser test client                     | Flutter build and end-to-end mobile hardware validation                 |

“Implemented” describes code in this prototype. It does not mean the behavior has passed physical radio, battery, button, LED, or multi-node testing. Those checks require the actual hardware.

## A few useful terms

- **LoRa:** the low-power radio link between nearby devices. It is not Wi-Fi and it carries much less data than a phone internet connection.
- **Mesh:** nodes can forward packets for one another, allowing a message to travel beyond a single radio link when relays are present.
- **Group channel:** messages use a shared key for a group of radios. A node without that key cannot read the encrypted group text.
- **Advert:** a public signed announcement of a MeshCore node identity; it is different from an encrypted Uni-Fi presence message.
- **Acknowledgement:** in Uni-Fi, a person’s reply to a particular distress event. It is not the same as the radio’s internal packet handling.

## One important limitation

The group key is shared by provisioned devices. The current message’s name and node fields help the application identify a claimed sender, but do not prove which person created it. Uni-Fi also inherits the current MeshCore group-encryption design. These limits, exact packet layouts, code paths, and validation results are described in the [technical walkthrough](#).

Uni-Fi is a coordination aid, not a guaranteed emergency-service connection.